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23DEA3202: DATA EXPLORATION AND PREPARATION

Semester	:	**B. Tech (VI semester)**
Name of the Student	:	M. Srija
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Major Project Title	:	Air Pollution Level Pattern Analysis
Domain Name	:	Smart cities and IOT
Base Paper Title	:	Analysis of Air Quality Patterns Using Environmental Monitoring Data

Abstract:
Air pollution has emerged as a major challenge in rapidly growing urban areas due to increased industrial activity, vehicular emissions, and population density. With the adoption of smart city initiatives, IoT-based air quality monitoring systems generate large volumes of environmental data that can be effectively analyzed to understand pollution behavior. This project presents an exploratory analysis of air pollution data collected from smart city monitoring systems to study spatial and temporal variations in air quality.

The analysis focuses on commonly monitored pollutants such as PM2.5, PM10, NO_2, SO_2, and CO across different locations and time periods. Exploratory Data Analysis (EDA) techniques are employed to examine daily, monthly, and seasonal pollution trends, identify peak pollution periods, and compare pollution levels across urban regions. Various data visualization techniques, including line plots, bar charts, and heatmaps, are used to clearly interpret and communicate the observed patterns.

The findings of this study reveal significant variations in pollution levels based on time and location, indicating the influence of urban activities and environmental factors. The insights obtained from this analysis can support data-driven decision-making for urban planning and pollution control, thereby contributing to improved air quality management in smart cities.

Keywords:
Smart Cities, Air Quality Analysis, Data Exploration, Environmental Monitoring

Recommended / Not Recommended (NR)

If NR, Remarks: _____

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